FLEET TEAM DIGEST

737MAX-FTD-32-18004

Issue Title : Low speed brake shudder (gear walk) on 737MAX Airplanes

Airplane Model	ATA	Minor Model(s)
737MAX	3241-41	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
08/27/2018	08/27/2018	06/19/2018

Estimated Completion Date	Next Update Date

Applicability

All 737MAX models.

Description

Operators have reported that while taxiing at very slow speeds, airplane shudder/vibration would occur while applying brakes.

The vibration is known as "Gear Walk." Gear walk, sometimes called "Brake Shudder," is a known and temporary phenomenon which can occur when performing low speed taxi stops with warm or hot brakes.

Background

Gear walk has occasionally been reported during taxi stops before airplane delivery or in service. Gear walk typically occurs at low speeds with warm or hot brakes and with light brake pedal application. Gear walk is a low-frequency vibration mode in which the brake and tire response couples with the natural frequency of the main landing gear. While the vibration may be noticeable in the cabin, there have been no reports of airplane damage due to this vibration mode.

With carbon brakes, which the 737MAX is equipped with, vibration is more likely to occur when the brake heat sinks (rotor and stator disks) are new. After a small number of moderately aggressive landing stops, the rubbing surfaces develop a friction film that stabilizes the brake friction coefficient, thereby reducing the likelihood of vibration. The brake friction coefficient can be higher than normal until this friction film develops. When brakes are applied gently at low speeds, a stick-slip condition can occur as the rubbing surfaces alternate between grabbing (sticking) due to high friction coefficient and rotating (slipping) under the low applied brake pressure. This condition is more likely to occur when the brakes are warm or hot because increased temperature tends to increase the brake friction coefficient.

Status

Closed

Interim Action

N/A

Final Action

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No final action required

Operator Action

If an airplane experiences a gear walk condition, it is our experience that the problem will diminish after several landing stops are performed with moderately aggressive braking. The problem may also be avoided by applying the brakes more firmly at low speeds. As a result, Boeing recommends that no maintenance actions be taken, unless the gear walk continues at an unacceptable level for passenger comfort/perception. If it is determined that a maintenance action is necessary, the brakes may be replaced at a convenient opportunity.

Related Categories

- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.

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